

COURSE: WEB TECHNOLOGIES (C2111C3) | UNIT 1 (4 CONTACT HOURS)

Unit 1: World Wide Web Architecture

A Deep-Dive Engineering Analysis of Protocols, Server Architectures, Dynamic IP, Web 2.0 & Design Standards

ITM (SLS) Baroda University, Vadodara | Faculty Lecture Series

Department of Computer Science & Engineering

Unit 1 Lecture Roadmap (4 Hours)

01

Hour 1: Foundations

Topics:

World Wide Web Concept,
Internet vs. WWW Architecture,
URIs/URLs, W3C Standards &
Hypermedia History.

02

Hour 2: Protocols & Clients

Topics:

HTTP/1.1 & HTTP/2 Mechanics,
Request/Response Headers,
Status Codes, Browser
Rendering & Web Servers.

03

Hour 3: Infrastructure

Topics:

Web 2.0 Paradigm
(AJAX/REST), Linux Server
Setup (Apache/Nginx), Virtual
Hosts, Dynamic IP & Dynamic
DNS.

04

Hour 4: Web Design

Topics:

Effective UI/UX Web Design,
Visual Hierarchy, Responsive
Web Design (RWD),
Accessibility (WCAG) &
Performance.

Introduction to World Wide Web (www)

CORE CONCEPT

System of Interconnected Pages

The World Wide Web (WWW), often called the Web, is a system of interconnected web pages and information that you can access using the Internet.

It was created to help people share and find information easily, using links that connect different pages together.

KEY CHARACTERISTICS

Global Interaction: Allows users to browse websites, watch videos, shop online, and connect with others worldwide.

The W3 Pool: All public websites accessible through the internet on any local device are collectively known as W3.

Hyperlink Navigation: Users discover further information by navigating interlinked pages and documents.

Rich Formats: Information is represented in diverse text, picture, audio, or video formats.



Global Connectivity Fact

Today, the Web connects *over 63% of the world's population*, making it one of the most powerful tools for communication and information sharing.

History of World Wide Web (WWW)

CORE ORIGIN

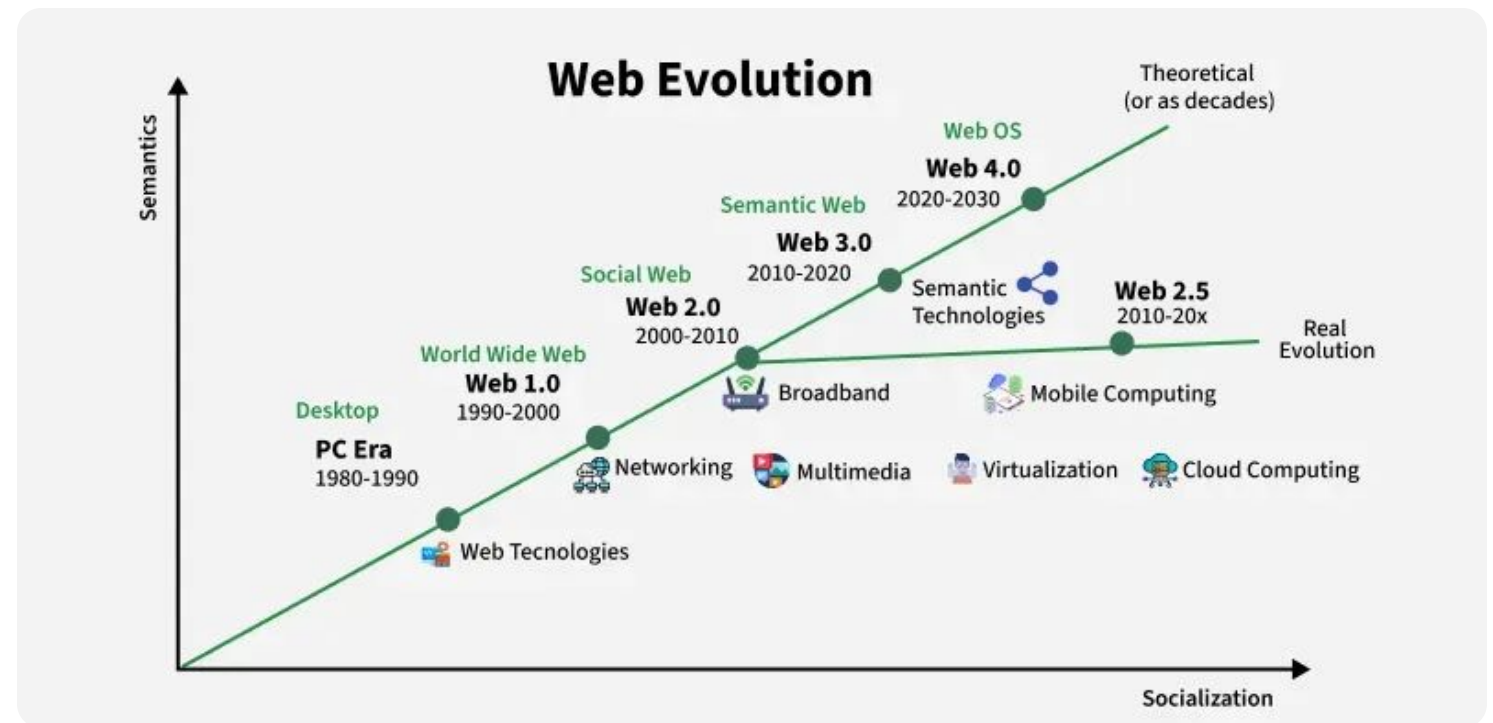
CERN & W3C (1989)

The World Wide Web was created by Tim Berners-Lee in 1989 as a project to help researchers at CERN collaborate and share information seamlessly.

To ensure its open growth, the World Wide Web Consortium (W3C) was founded to develop common standards and specifications for the web.

ROADMAP

How the Web Evolves



History of World Wide Web (WWW) (Contd.)

ROADMAP

The Evolution of the Web

Web 1.0 (1990–2000)

Introduced static websites that primarily allowed one-way information sharing.

Web 2.0 (2000–2010)

Brought interactive, social platforms enabling user-generated content and global community connection.

Web 3.0 (2010–2020)

Focused on the semantic web, allowing machines to understand data and provide deeply personalized experiences.

Web 4.0 (2020–2030)

Expected to be a fully intelligent, decentralized ecosystem powered by AI and integrated web operating systems.

STATISTICS & INSIGHTS

Some Facts About the Web

The First Website is Still Live

The historical first website is still fully active. You can visit it anytime at <http://info.cern.ch/>

1.5 Billion+ Active Websites

The internet ecosystem currently hosts more than 1.5 billion unique websites, and this massive number grows daily.

Chrome's Massive Dominance

Google Chrome has captured the global market, used by more than 60% of people browsing resources online.

Truly Global and Multilingual

The Web is globally inclusive, officially operating in over 150 diverse languages to connect users worldwide.

Concept of World Wide Web (www)

Introduction

The Web Browser

A software interface used to access and display hyperlinked resources on the World Wide Web, translating code into accessible multimedia.

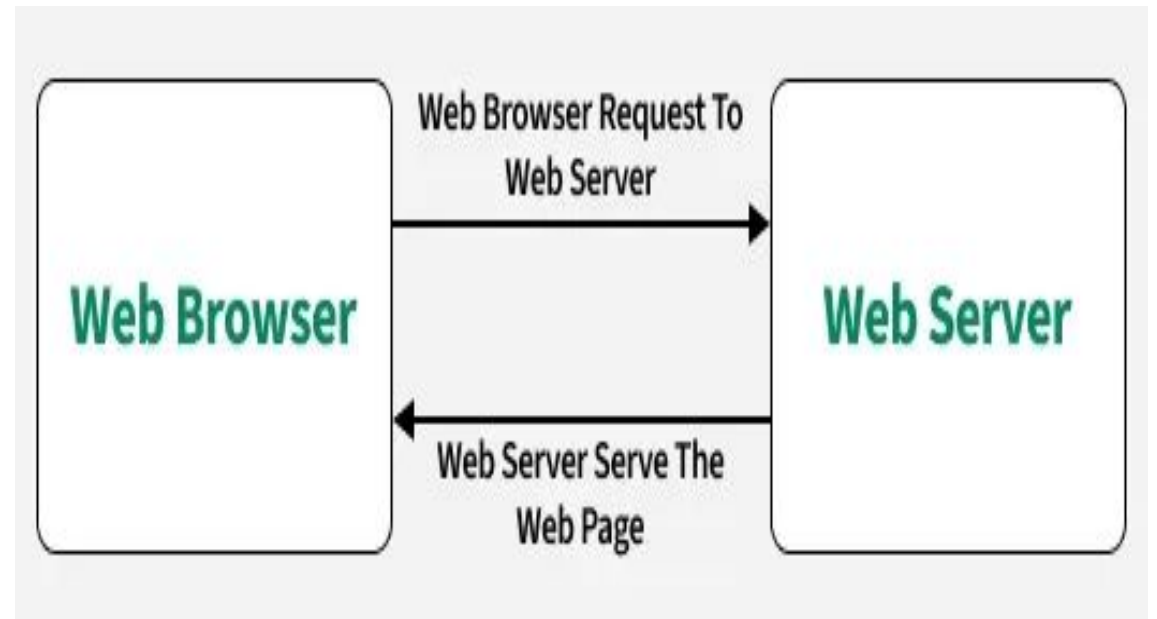
Key Functions & Capabilities

- **Resource Rendering:** Displays text, data, pictures, animation, and video seamlessly.
- **Multi-tasking Utility:** Conducts searches, transfers files, and handles mailing services.
- **Common Software:** Includes popular browsers like Google Chrome, Opera Mini, and Internet Explorer.

Architecture

Client-Server Model

The web operates on client-server architecture: browsers request files, and servers return them to the client.



Internet Infrastructure vs. WWW

The Internet

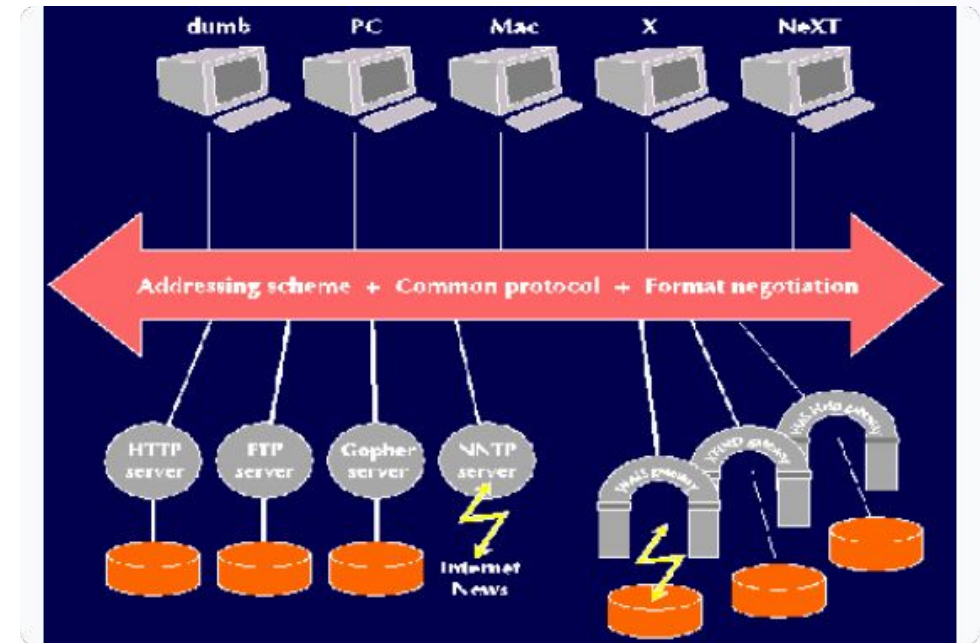
Hardware/Network Layer

A globally decentralized network of networks (autonomous systems) operating on TCP/IP. Handles physical routers, fiber optics, packet switching, IP routing, and low-level transport protocols (TCP/UDP).

World Wide Web

Application/Data Layer

An information system running on top of the Internet. Invented by Sir Tim Berners-Lee at CERN (1989). Utilizes Hypertext, Uniform Resource Identifiers (URIs), and HTTP to interlink web documents worldwide.



Key Takeaway for CSE Engineers

The Internet provides the *infrastructure* (roads), while the WWW provides the *service* (vehicles and traffic running on the roads).

HTTP Protocol & Header Anatomy

HTTP Request Structure

Client Request

1. Request Line

Method (GET, POST, PUT, DELETE),
Request-URI (/api/data), **Protocol Version** (HTTP/1.1)

2. Headers

Key-value metadata passed by client:

- **Host:** www.itm.ac.in (Mandatory in HTTP/1.1)
- **User-Agent:** Mozilla/5.0... (Browser identity)
- **Accept:** text/html, application/json

3. Body

Payload for POST/PUT forms/JSON.

HTTP Response Structure

Server Response

1. Status Line

Version, Status Code & Reason (e.g., HTTP/1.1 200 OK or 404 Not Found)

2. Headers

Server metadata returned to client:

- **Content-Type:** text/html; charset=utf-8
- **Set-Cookie:** session_id=xyz123
- **Cache-Control:** max-age=3600

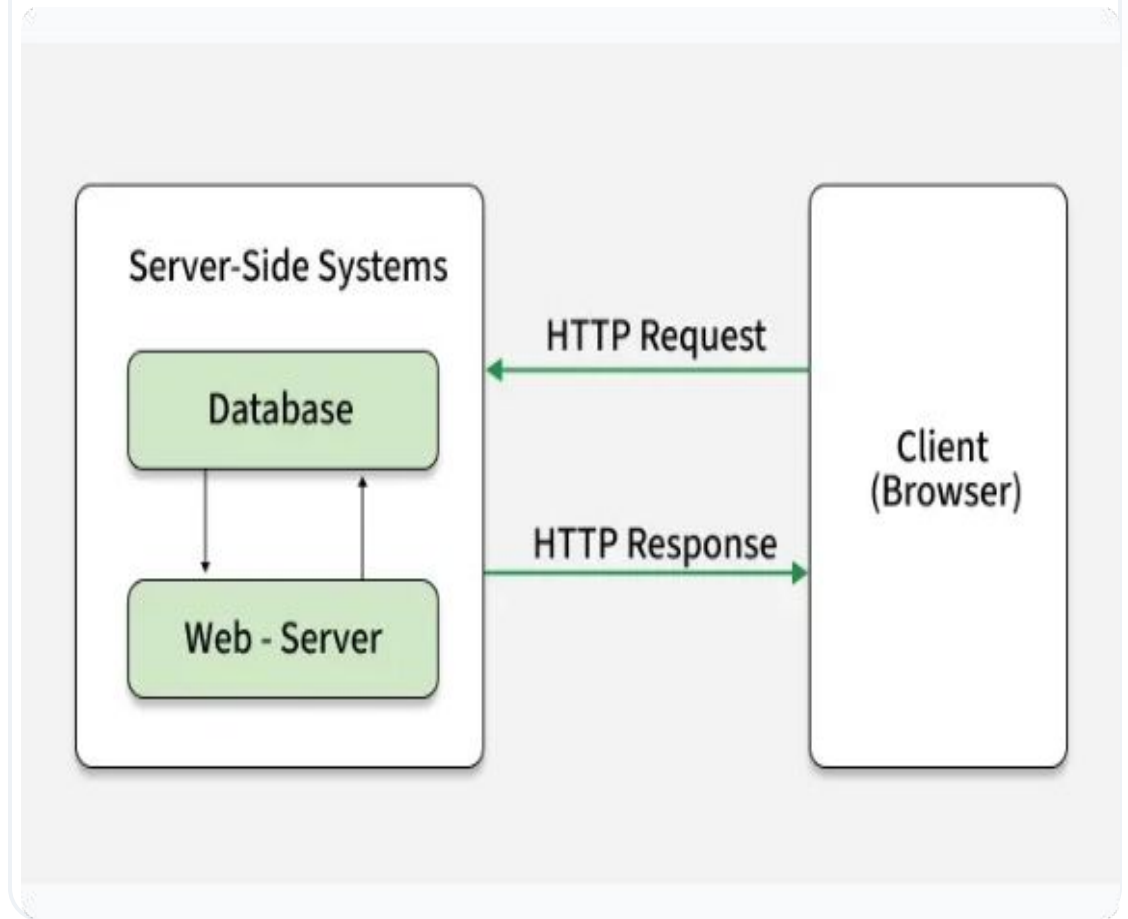
3. Body

HTML document, CSS file, or API JSON payload.

HTTP Request-Response Lifecycle

- 1 Domain Name Resolution (DNS)**
Browser converts `http://example.com` to IP via Local DNS Cache Recursive Resolver Root/TLD Authoritative DNS.
- 2 TCP 3-Way Handshake**
Client sends SYN, Server responds with SYN-ACK, Client returns ACK (establishes reliable socket connection on port 80 or SSL 443).
- 3 HTTP Transmission**
Client transmits HTTP Request. Server processes routing logic, database queries, and returns HTTP Response.
- 4 Connection Teardown / Keep-Alive**
Socket closes or stays open for subsequent HTTP requests via persistent connections.

Request-Response Architecture



Understanding a Uniform Resource Locator (URL)

`https://www.itmbu.ac.in`

1. SCHEME / PROTOCOL

https://

Hypertext Transfer Protocol Secure

Defines the protocol used to transmit data between your browser and the server.

The "Secure" aspect:

The **s** indicates an encrypted SSL/TLS connection, protecting user data from interception and tampering.

2. SUBDOMAIN

www

World Wide Web

A subdivision of the main domain name.

Traditionally designates the primary web server hosting public pages. While modern sites can load without it, it remains a standard convention.

3. SECOND-LEVEL DOMAIN

itmbu

Organization Identifier

Institute of Technology and Management Baroda University.

The unique name chosen to represent its brand or identity on the internet, distinguishing it from all other entities.

4. TOP-LEVEL DOMAIN

.ac.in

Compound TLD Structure

.ac (Academic):

Reserved for academic and educational institutions in India.

.in (India): Country Code TLD (ccTLD) designated for India, managed by NIXI.

The combination instantly validates the site as an accredited Indian educational institution.

URI vs. URL

1. URI (Uniform Resource Identifier)

An overarching umbrella term for characters used to **identify** a resource on the internet. It gives a unique identity without necessarily showing how to access it.

Two main sub-categories:

- **URLs:** Identifies a resource by its **location**.
- **URNs:** Identifies a resource by its unique **name**, regardless of location.
`urn:isbn:978-0-395-36341-6`

2. URL (Uniform Resource Locator)

A specific type of URI that not only identifies the resource but provides explicit instructions on **how to locate and access it** by including a protocol.

Key Components & Example:

- Requires a scheme/protocol (like `https://`, `ftp://`).
- `https://www.itmbu.ac.in` is a URL because it details the exact protocol needed to reach the destination.

The Real-World Analogy

Think of a **URI** as a person's **identity**, and a **URL** as their **home address**.

URI (The Identity): "John Doe" (Tells you *who* the resource is, but not how to get to him). **URL (The Locator):** "123 Main Street, Vadodara, Gujarat" (Tells you exactly *how to reach him*).

URI vs. URL (Contd.)

URI (Identifier)

Primary Purpose

To uniquely identify a resource.

Protocol Included?

Not always required (can just be a name).

Relationship

The parent category.

Example

`urn:isbn:0451450523`
(identifies a specific book)

URL (Locator)

Primary Purpose

To identify a resource *and* locate it.

Protocol Included?

Always required (e.g., `http://`, `ftp://`).

Relationship

A sub-category of a URI.

Example

`https://www.google.com`
(tells you where to go)

Domain Name Structure & Purpose

IP Address vs. Domain Name

How Computers are Located

Every internet-connected device has a public IP address:

- **IPv4:** e.g., 192.0.2.172
- **IPv6:** e.g.,
2001:db8:8b73::8a2e:370:1337

The Human-Readable Solution

IP addresses are highly difficult for humans to memorize and frequently change over time.

To bridge this gap, we use memorable, human-readable **Domain Names** to represent complex server IP addresses.

Structure of Domain Names

Domain names consist of multiple distinct labels separated by dots.

Important: Domain structures are parsed and read strictly from right to left.

developer.mozilla.org

The diagram illustrates the structure of the domain 'developer.mozilla.org'. It shows three labels separated by dots. The first label, 'developer', is grouped by a bracket and labeled 'label 2'. The second label, 'mozilla', is grouped by a bracket and labeled 'label 1'. The third label, 'org', is grouped by a bracket and labeled 'TLD'.

TLD (Top-Level Domain)

TLD Purpose & Specifications

General Purpose

TLDs tell users the general purpose of the service behind the domain name. Generic TLDs like **.com**, **.org**, and **.net** do not require users to meet any strict criteria.

Technical Limits

- Can contain special as well as latin characters.
- Maximum length of **63 characters** (most are around 2-3).

The full list of official TLDs is actively **maintained by ICANN.**

Enforced Policies & Restrictions

Some TLDs enforce strict policies so the intent and background of the service is transparent to the end-user:

Local / Country TLDs

TLDs like **.in**, **.fr**, or **.us** can require the service to be provided in a given language or hosted within that specific country.

Government Restricted

TLDs containing **.gov** are exclusively reserved for government departments and official authorities.

Educational Institutions

The **.edu** & **.ac** TLD is strictly restricted for accredited educational and academic institutions.

Label (or Component)

Label Specifications

What is a Label?

The labels are the individual components that follow the TLD (Top-Level Domain) in a domain name structure.

Technical Constraints

- **Length:** Anywhere from 1 to 63 characters in length.
- **Case Sensitivity:** Labels are entirely case-insensitive.
- **Characters:** Only letters (A-Z), digits (0-9), and the hyphen (-).
- **Hyphen Rule:** A hyphen may not be the first or last character.

Valid Examples

a

97

hello-strange-person-16-how-are-you

SLDs & Subdomains

Secondary Level Domain (SLD)

The label located directly before the TLD is referred to as the **Secondary Level Domain (SLD)**.

Multi-Label Structures

A domain name is not limited to a rigid 3-label standard. It can have many visual components separated by dots.

`informatics.ed.ac.uk` is a perfectly valid multi-label domain name.

Subdomains

For any base domain you control (e.g., `mozilla.org`), you can dynamically delegate "subdomains" with unique content:

`developer.mozilla.org`

`support.mozilla.org`

`bugzilla.mozilla.org`

Buying a Domain Name

Who Owns a Domain Name?

You cannot "buy" a domain name permanently.

This system ensures that unused domain names eventually become available to be used again by others. If domains were permanently owned, the web would quickly fill up with unused, locked names.

Paying for the Right to Use

Instead, you pay for the **right to use** a domain name for one or more years. You have priority to renew your right over other applications, but you never truly own the domain name.

How Domains Are Managed

Domain Registrars & Registries

Companies called **registrars** use registries to track technical and administrative information connecting you directly to your domain name.

Exceptions to the Rule

For some domains, a standard registrar might not be in charge of keeping track [cite: 16].

Example Case:

Every domain name under **.fire** is managed directly by Amazon.

Web Browsers vs. Web Servers

Web Browser (User Agent Engine)

Core Objective

Fetch remote web assets, parse code, construct DOM/CSSOM trees, render layout, and execute client scripts.

Key Internal Components

- **Rendering Engine:** Blink (Chrome), Gecko (Firefox), WebKit (Safari).
- **JavaScript Engine:** V8 (Chrome), SpiderMonkey (Firefox).
- **Network & Storage Layer:** HTTP cache, LocalStorage, Cookies, TLS stack.

Web Server (Server Daemon)

Core Objective

Listen on network sockets (0.0.0.0:80/443), accept HTTP requests, evaluate access controls, serve static files or route to application runtimes.

Architectural Models

- **Process/Thread per Connection:** Apache HTTP Server (MPM Prefork/Worker). High memory usage under concurrent load.
- **Asynchronous Event-Driven:** Nginx / Node.js. Non-blocking I/O loop handling 10k+ concurrent connections efficiently.

Difference Between IPv4 and IPv6

Protocol Overview

The Internet Protocol (IP) assigns a unique address to each device, enabling communication over the internet.

IPv4 (The Earlier Version)

Uses **32-bit** addresses, providing about **4.3 billion** unique addresses, which are now nearly exhausted.

IPv6 (The Newer Version)

Uses **128-bit** addresses, offering a vastly larger address space of **340 undecillion** to support future growth.



Drawbacks of IPv4

01

Scalability & Capacity

- **Limited Address Space:**
The 32-bit IPv4 address space is insufficient for the rapidly growing number of connected devices worldwide.
- **Broadcast Overhead:**
Broadcast traffic increases unnecessary network load and degrades overall system performance.

02

Routing & Efficiency

- **Complex Configuration:**
Heavy reliance on manual setup or DHCP often leads to human errors and configuration mismatches.
- **Inefficient Routing:**
Complex, variable-length headers increase processing overhead at every routing hop.
- **Router Fragmentation:**
Packet fragmentation managed by routers slows down throughput and risks data packet loss.

03

Security & Performance

- **No Built-in Security:**
Security features (IPsec) are not natively integrated and always require complex, external add-ons.
- **Limited QoS Support:**
Poor native traffic prioritization impairs real-time apps like high-definition video and voice.

Benefits of IPv6 over IPv4

Key Advantages

IPv6 offers massive improvements in scalability, performance, and built-in security features over IPv4.

- **Larger Address Space:** Uses 128-bit addresses (vs. 32-bit) to support billions of new devices.
- **Improved Security:** Built-in data authentication and encryption protocol standards (IPsec).
- **Simpler Header:** Streamlined fixed-size header format that increases routing speed and efficiency.
- **Better QoS Support:** Native handling of quality-of-service parameters for optimized streaming.
- **Mobile-Friendly:** Seamless and highly secure support optimized for mobile connections.

Transition Strategies

Coexistence techniques that allow a smooth, gradual transition from IPv4 to IPv6 architectures.

- **Dual Stacking:** Devices run both IPv4 and IPv6 network stacks simultaneously to ensure compatible communications.
- **Tunneling:** Encapsulates IPv6 data packets inside older IPv4 systems to seamlessly traverse non-upgraded networks.
- **Network Address Translation (NAT):** Helps completely disparate IP versions understand each other by dynamically translating address headers.

IPv4 vs IPv6 Comparison

Feature	IPv4	IPv6
Address Size	Uses a 32-bit IP address.	Uses a 128-bit IP address.
Notation	Uses decimal dot-separated notation.	Uses hexadecimal colon-separated notation.
Address Space	Limited space of about 4.3 billion addresses.	Extremely large space for future growth.
Configuration	Supports manual configuration and DHCP.	Supports SLAAC, DHCPv6, and manual setup.
Connectivity (NAT)	Often affected due to NAT.	End-to-end connectivity is restored without NAT.
IPsec Security	Optional.	Built into the protocol design.
Fragmentation	Performed by both sender and routers.	Performed only by the sender.
Packet Flow ID	Does not support flow-based identification.	Uses a Flow Label field.
Header Checksum	Included.	Does not include a header checksum.
Transmission	Supports broadcast communication.	Uses multicast and anycast instead.
Header Size	Variable (20–60 bytes).	Fixed at 40 bytes.
Address Classes	Uses classes (A, B, C, D, and E).	Does not use address classes.
Subnetting	Supports Variable Length Subnet Masking (VLSM).	Uses prefix-based addressing.
Example Address	66.94.29.13	2001:db8::1

Evolution to Web 2.0 Technologies



TECHNOLOGY 01 **AJAX & Async Processing**

Introduction of XMLHttpRequest and modern fetch() API allowed web applications to request data from servers in the background without requiring full page reloads, enabling rich Single Page Applications (SPAs).



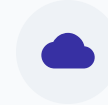
TECHNOLOGY 02 **User-Generated Content**

Shift from read-only static websites to participatory platforms (Social Networks, Wikis, Blogs) where end users actively publish, comment, and collaborate.



TECHNOLOGY 03 **RESTful APIs & Services**

Standardization of lightweight JSON data exchanges over HTTP REST APIs, replacing heavy SOAP XML protocols and allowing seamless third-party web mashups.



TECHNOLOGY 04 **Software as a Service**

Applications shifted from desktop installation binaries to cloud-hosted browser applications (e.g., Google Docs, Figma, Office 365) running seamlessly everywhere.

Dynamic IP & DDNS Resolution



Concept 01

Static IP Address

Permanently assigned IPv4/IPv6 address that remains constant. Essential for production web servers, database clusters, and DNS authoritative servers. Highly expensive.



Concept 02

Dynamic IP Address

Temporarily assigned IP address allocated by an ISP via **DHCP lease cycles**. IP changes periodically upon router reboot or lease expiration.



The Problem

Resolution Failure

Standard DNS A-records require static IP mapping. When a home or small business server's Dynamic IP changes, the domain lookup immediately fails.



The Solution

Dynamic DNS (DDNS)

A lightweight daemon runs on the server. Whenever the WAN IP changes, it sends an authenticated API request to update the DNS A-record dynamically within seconds.

IP Address Definition & Explanation



Core Definition

An Internet Protocol (IP) address is a unique identifying numeric label assigned to every device connected to the internet or local network, enabling essential communication and spatial location tracking.



Versions & Types

Features two key standards: IPv4 (older, handles 4 billion addresses) and IPv6 (modern, supports trillions). IP addresses are classified into public, private, static, and dynamic types.



Device Connectivity

Every connected hardware—laptops, IoT systems, smart appliances, or even toys—requires an IP address to efficiently route and transfer payloads between distinct networks.

How does an IP address work?



Locate & Retrieve

An IP address serves as a locator, helping your device find and retrieve whatever data or content you are accessing on the internet.



Identify & Map

It identifies host networks or device locations. Based on math, addresses are allocated by Internet Assigned Numbers Authority (IANA) and range from 0.0.0.0 to 255.255.255.255.



Connect Devices

Through mathematical assignment, the unique identification establishes a direct and functional connection to any online destination.

Public IP address



External Connection

Applies to the main device (typically the router) connecting a home or business network to the Internet Service Provider (ISP). All local devices communicate externally using this single public IP address.



Crucial Use Cases

Knowing an external-facing IP address is crucial for configuring ports. It enables smooth online gaming, hosting email and web servers, streaming media, and establishing secure remote connections.

Private IP address



Local Routing

Assigned by local intranets or ISPs. The home or office router manages these internal-facing addresses, mapping devices from private to public IP addresses.



Address Reuse

Private addresses are reused across multiple networks, preserving valuable IPv4 space and extending addressability beyond the 4,294,967,296 (2^{32}) limit.



IPv6 & ULA

In IPv6, devices get unique identifiers from the ISP or network organization. Private addressing is also supported and referred to as Unique Local Addressing (ULA).

Static IP Address



Core Concept

An IP address that a person manually configures and fixes to their device's network is referred to as a **static IP address**. A static IP address cannot be changed automatically.



Address Types

All public and private addresses are defined as ***static*** or ***dynamic***, establishing the core framework for network device identification.



ISP Allocation

An **internet service provider** may assign a static IP address to a user account. The same IP address will be assigned to that user for every session.

Dynamic IP Address



DHCP Server

The **Dynamic Host Configuration Protocol (DHCP)** assigns the distribution of this dynamic set of IP addresses. The DHCP can be the router that provides IP addresses to networks across a home or an organization.



Auto Assignment

A *dynamic IP address* is automatically assigned to a network when a router is set up. Each time a user logs into the network, a fresh IP address is assigned from the pool of available, currently unassigned IP addresses.



Address Cycling

A user may randomly cycle through several IP addresses across multiple sessions. This seamless rotation relies on a pool of currently unassigned IP addresses to establish flexible device identification.

Four Types of IP Addresses

Four types of IP Addresses

Public



Private



Static



Dynamic



Four Types of IP Addresses



Public IP

An external-facing address assigned to your router by the ISP, used to communicate with the wider internet.



Private IP

An internal address assigned by your router, allowing devices within your local network to communicate securely.



Static IP

A permanent, unchanging address configured for servers or devices requiring consistent connection points.



Dynamic IP

A temporary address assigned automatically on a rotating basis, standard for most consumer internet devices.

What Is IPv4?



Core Protocol

IPv4 is the fourth version of the Internet Protocol. It is a core standard-based method used to interconnect networks globally. It was first deployed on the Atlantic Packet Satellite Network (SATNET) in 1982.



Current Usage

Despite the transition to IPv6, IPv4 remains the primary router of most internet traffic today. It continues to be actively assigned to and utilized by computers across global networks.



Address Format

Uses 32-bit binary numbers. It consists of four sets of numbers ranging from 0 to 255 (each representing an eight-digit binary number), separated by period points (e.g., 10.0.0.1 or 197.0.23.10).

Reserved IP Address Classes

Some IP addresses are reserved by the Internet Assigned Numbers Authority (IANA) for specific purposes on the Transmission Control Protocol/Internet Protocol (TCP/IP) suite. Four key reserved IPv4 classes include:



0.0.0.0

Default Network

A non-routable meta-address used to designate an invalid, non-applicable, or unknown network target in IPv4.



127.0.0.1

Loopback Address

An address a computer uses to identify itself and test local network services, regardless of its assigned IP.



169.254.0.1–254

Link-Local IP

A range automatically assigned to clients when they are unable to retrieve an IP address from a DHCP server.



255.255.255.255

Broadcast IP

A dedicated address used to transmit messages to every single connected machine on a local network segment.

Understanding Subnet Classes

Further reserved IP addresses are allocated for subnet classes. Subnetworks are smaller networks that connect to a larger network via a router. By assigning its own local IP address scheme, devices inside a subnet can communicate directly with each other without routing unnecessary data through the wider internet.

IP Address Classes (Contd.)

The router on a TCP/IP network can be configured to ensure it recognizes subnets, then route the traffic onto the appropriate network. IP addresses are reserved for the following subnets:



Class A

Common Subnet

IP addresses
between
10.0.0.0 to
10.255.255.255



Class B

Common Subnet

IP addresses
between
172.16.0.0 to
172.31.255.255



Class C

Common Subnet

IP addresses
between
192.186.0.0 to
192.168.255.255



Class D

Multicast usage

IP addresses
between
224.0.0.0 to
239.255.255.25
5



Class E

Experimental

IP addresses
between
240.0.0.0 to
254.255.255.25
4

Subnet Allocation & Standards Guidance

Class A, B, and C are the most commonly used ranges in the creation of subnets. Addresses within the multicast range (**Class D**) have specific usage rules outlined in the Internet Engineering Task Force (IETF) guidelines, while the release of **Class E** addresses for public use was the cause of plenty of debate before the IPv6 standard was introduced.

Internet Addresses and Subnets



Global Allocation

IANA & ISP Blocks

The IANA reserves specific IP address blocks for commercial organizations, government departments, and ISPs.

When a user connects, their ISP assigns them an address from these blocks. If online from one computer, they use that direct ISP-assigned address.



Home Routers

Shared Connections

Most homes now use routers that share a single network connection with multiple local devices.

The ISP assigns the IP address directly to the router, and a private subnet is then created for all computers that connect to it.



Network & Nodes

Subnet Masking

IP addresses within a subnet contain both a network ID and a node ID (host). Computers separate these using a subnet mask.

The mask filters the IP designation to determine the optimal layout based on required nodes or subnets.



Reserved IPs

Boundaries

Every subnet has boundaries. The very first IP address in the block is strictly reserved for identifying the subnet itself.

The final address in the subnet range is reserved as the broadcast address, used to send data to all systems on that subnet.

IPv4 vs IPv6



IPv4 Limitations

Exhausted Capacity

IPv4 has not been able to cope with the massive explosion in the quantity and range of devices beyond simply mobile phones, desktop computers, and laptops.

The original IP address format was not able to handle the number of IP addresses being created.



IPv6 Solution

Next-Gen Standard

To address this problem, IPv6 was introduced. This new standard operates a hexadecimal format that means billions of unique IP addresses can now be created.

As a result, the IPv4 system that could support up to around 4.3 billion unique numbers has been replaced by an alternative that, theoretically, offers unlimited IP addresses.



IPv6 Structure

Hexadecimal Format

An IPv6 IP address consists of eight groups that contain four hexadecimal digits, which use 16 distinct symbols of 0 to 9 followed by A to F to represent values of 10 to 15.

How Do I Locate My IP Address?



Windows PC Command Prompt

1. Type `cmd` into the search tab and press Enter.
2. In the command prompt, type `ipconfig` to find your IP.



Apple Mac System Settings

1. Go to the Apple menu and select **System Preferences**.
2. Click on **Network** to view your active network details.



Mobile Devices Settings & Wi-Fi

1. Go to **Settings** and open the **Wi-Fi / Network** menu.
2. Locate your network to find the IP listed under **Advanced**.

IP Address vs. MAC Address



IP Address

Network Connection Identifier

An IP address identifies the connection of a device to a network. It is provided by the Internet Service Provider (ISP) and is necessary to route network packets to their destination.

- **Format & Layer:** 4-byte (IPv4) or 16-byte (IPv6) address operating in the Network Layer.
- **Dynamic Nature:** Addresses typically change dynamically rather than remaining permanent.
- **Visibility:** Can be easily found by third parties on the internet.



MAC Address

Physical Device Identifier

A MAC address is the physical address of a device. It is assigned and provided directly by the hardware manufacturer of the network card or router.

- **Format & Layer:** A 6-byte hexadecimal address operating in the Data Link Layer.
- **Static Nature:** Addresses are static and remain bound permanently to the hardware.
- **Visibility:** Highly private; third parties cannot see it unless they are on your Local Area Network (LAN).

What are security threats related to IP addresses?



Threat Overview

Deception & Surveillance

A variety of security threats are related to IP addresses.

Cybercriminals can deceive devices to either reveal your IP address and pretend they are you, or stalk it to track activity and take advantage.

Key Risks:

Online stalking and social engineering are the two leading security threats existing for IP addresses.



Geographical Location Tracking

Allows a cybercriminal to use your IP address to track your physical location and narrow down your precise area.



Targeted DDoS Attacks

Cybercriminals can use your IP address to target your home network and launch a devastating Distributed Denial of Service (DDoS) attack.



Illegal Content Downloads

Malicious actors can hijack or use your IP address to download illegal content, falsely implicating you in illicit activities.

5 ways to protect your IP address

There are multiple ways to protect your IP address from cybercriminals. Some of these options include:



VPN Protection

Use a VPN to encrypt your internet traffic and mask your true location.



Proxy Server

Make use of a proxy server to act as an intermediary for your web requests.



Dynamic IP

Have your ISP make use of dynamic IP addresses that change periodically.



NAT Firewall

Employ a NAT(Network Address Translation) firewall to hide your private IP address from external networks.



Modem Reset

Resetting your modem may change your IP address when a new connection is established.

IP Address FAQs



What is the use of an IP address?

An IP address identifies every device connected to the internet. This enables computers, mobile phones, and Internet-of-Things (IoT) devices to communicate over the internet and on local-area networks (LANs).



What happens if someone has your IP?

The hacking risk is low, but your location can be narrowed down. Cybercriminals might identify your ISP for phishing, or trade IPs on the dark web.

Providers like Netflix and Hulu also read IP addresses to enforce regional content restrictions.



What are the two types of IP addresses?

Public IPs are used to connect to the external internet (usually the router), while Private IPs are assigned internally to devices like laptops, printers, and smart TVs.



What is the 192.168 IP address?

It marks the starting point of the private IP range, running through to 192.168.255.255. This address is not typically assigned to standard devices on a network.



What is a loopback IP address?

The loopback address (127.0.0.1) is used by a computer to identify itself locally, functioning independently of whether a network IP has been assigned.

Effective Web Design Principles



Visual Hierarchy

Guide user eye movement using scale, typographic weight, and visual contrast. Structure layouts around standard scanning patterns (F-Pattern for text, Z-Pattern for visual landing pages).



Navigation & 3-Click Rule

Structure clear site maps, breadcrumbs, and persistent header menus. Users should reach any piece of content within 3 intuitive clicks from the home page.



Accessibility (WCAG 2.1)

Comply with Web Content Accessibility Guidelines: ensure minimum 4.5:1 text color contrast ratio, provide image alt attributes, and ensure full keyboard navigation.

Web Design Challenges & RWD



Challenge 01

Responsive Web Design

Designing fluid layouts across smartphones (360px), tablets (768px), and 4K displays using CSS Media Queries, CSS Flexbox, and CSS Grid systems.



Challenge 02

Cross-Browser Compatibility

Managing layout engine differences (Chromium vs. Gecko vs. WebKit). Requires CSS resets, Autoprefixer for vendor prefixes, and feature detection via Modernizr.



Challenge 03

Performance & Speed

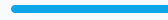
Optimizing Core Web Vitals (LCP, CLS, FID) affected by large assets. Mitigated via WebP compression, CSS/JS minification, CDNs, and async loading.



Challenge 04

Security Considerations

Ensuring robust security by preventing Cross-Site Scripting (XSS), mitigating Clickjacking risks, and strictly enforcing Content Security Policy (CSP) headers.



Web Design Fundamentals

Concepts of effective Web Design and Web Design Issues

SECTION **0**
1

Effective Concepts

The building blocks of a great user experience



**"Design is not just what it looks like
and feels like. Design is how it
works."**

— Steve Jobs

Core Concepts of UI/UX



User-Centricity

The best designs prioritize the needs of the end-user. Understanding the target audience allows developers to create intuitive flows that reduce friction.



Visual Hierarchy

By manipulating size, color, and spacing, designers can guide a user's eye organically through the interface, emphasizing the most important information first.



Accessibility (a11y)

Inclusive design ensures that websites are usable by everyone, including people with visual, motor, or cognitive disabilities, utilizing strict WCAG standards.

Responsive Web Design



Adaptability is Key

A responsive website fluidly adjusts its layout, typography, and images to fit any screen size—from a large desktop monitor to a small smartphone.

Using fluid grids and CSS media queries ensures that users have a seamless experience regardless of their device, which is foundational in modern web development.

Visual Hierarchy & The F-Pattern

- 01** Users rarely read web pages word-for-word; instead, they **scan** the content to find what they need quickly.
- 02** Eye-tracking studies reveal that users typically scan screens in an **F-shaped pattern**, focusing heavily on the top and left sides.
- 03** Important information (like logos, primary navigation, and key calls-to-action) must be placed along these main visual pathways.
- 04** Creating this clear hierarchy guides the user's attention seamlessly and prevents them from getting lost in a wall of text.



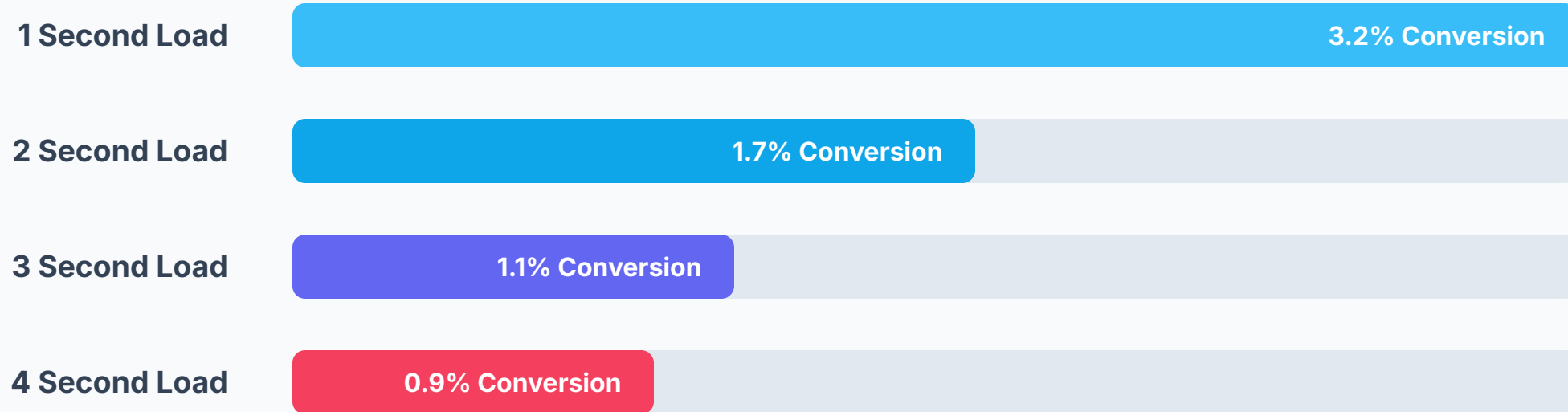
Section

02

Common Issues

Where web design goes wrong and how it impacts users

The Cost of Slow Load Times



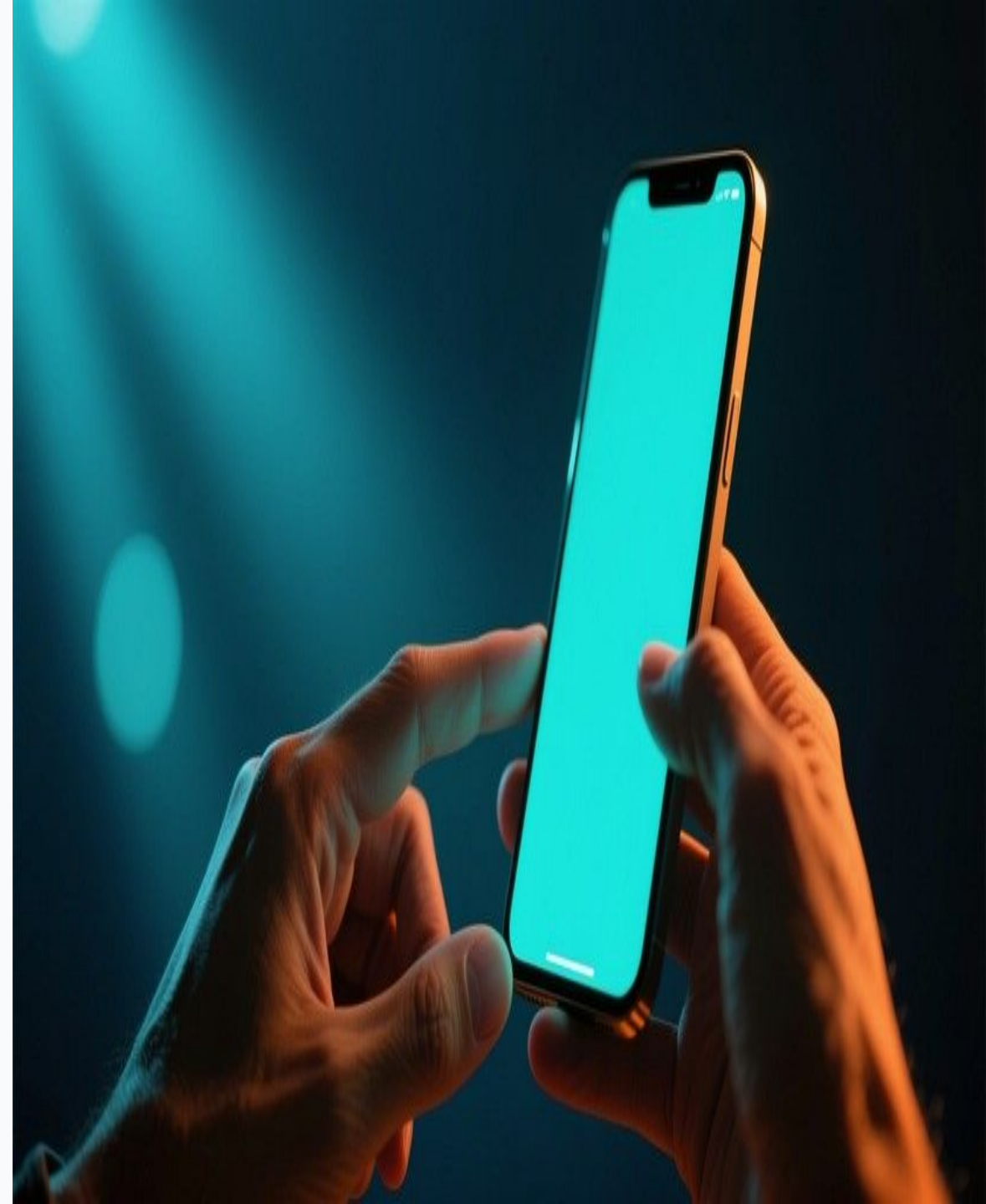
A website that loads in just 1 second sees significantly higher conversion rates compared to sites loading in 3 or 4 seconds. In the modern web, speed is absolutely critical for retention.

Mobile Unresponsiveness

The Mobile-First Imperative

Ignoring mobile users is a critical flaw. Websites that force users to pinch-and-zoom to read text create extreme usability friction.

This neglect leads to skyrocketing abandonment rates, poor brand perception, and significant penalties in modern search engine optimization (SEO) rankings.



Common Visual & Technical Pitfalls



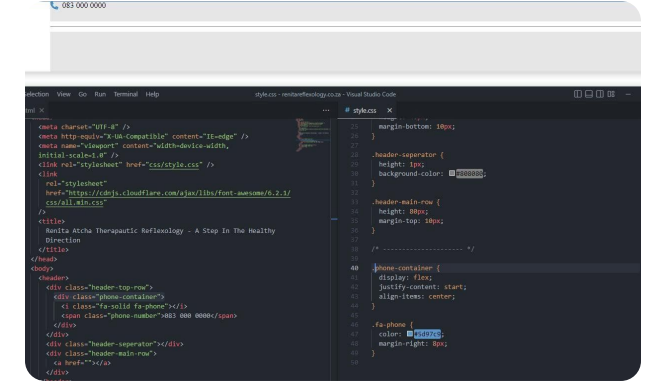
Cluttered Layouts

Overloading a page with text, images, and features causes cognitive overload. Whitespace is crucial.



Poor Accessibility

Low color contrast and small fonts make websites unusable for many people. Accessibility is not optional.



Technical Glitches

Broken links, outdated code, and server errors severely damage user trust and search engine indexing.

Best Practices Checklist



Mobile-First Approach

Always design and optimize for mobile devices first, as they account for the vast majority of global web traffic.



Prioritize Performance

Optimize images, minimize code, and leverage caching to ensure lightning-fast page loading speeds.



Clear Visual Hierarchy

Use typography, whitespace, and color strategically to effortlessly guide the user's eye to the most important elements.



Ensure Accessibility

Design inclusively. Provide alt text for images, maintain high contrast ratios, and test for keyboard-friendly navigation.

Linux Web Server Setup & Config

Apache & Nginx Deployment Steps



Step 01

Package Manager Installation

Execute `sudo apt update && sudo apt install apache2 nginx` on Debian/Ubuntu Linux.



Step 02

Directory & Permissions

Set web document root at `/var/www/html`. Assign ownership to server user via `chown`.



Step 03

Virtual Host Configuration

Create domain configuration files in `sites-available` to host multiple domains on a single IP address.



Step 04

Firewall & SSL Setup

Open HTTP/HTTPS ports using `ufw` and issue SSL certificates via Let's Encrypt Certbot utility.

Unit 1 Summary & Review Questions



Key Review Questions Exam Preparation Focus

- Q1.** Differentiate between the Internet and WWW with suitable examples. **(4 Marks)**
- Q2.** Explain the HTTP request-response header structure in detail. **(6 Marks)**
- Q3.** Describe the mechanism of Dynamic IP and how Dynamic DNS resolves IP shifts. **(4 Marks)**
- Q4.** What are the key features of Web 2.0? Explain AJAX. **(5 Marks)**



Lab & Next Lecture Practical Alignment



Lab Experiment Connection

The server setup and HTTP concepts from Unit 1 form the baseline for deploying your HTML/CSS web projects in **Lab Experiments P1 to P5**.



Next Lecture (Unit 2)

HTML5 Fundamentals — Document Type, Formatting Tags, Forms, HTML5 Media Elements & Semantic Tags.